

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5	Annexure A	Total Marks:25
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Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

1. Star vs. Bus Topology: Cost, Reliability, and Performance

Star and bus topologies represent two distinct approaches to local network deployment, differing fundamentally in central hardware reliance versus shared transmission lines.

Cost and Physical Cabling

- **Star Topology:** Involves higher initial capital expenditure due to mandatory central switches/hubs and independent cable runs from every individual endpoint.
- **Bus Topology:** Highly budget-friendly with lower cabling demands, relying on a single central backbone cable with short drop lines connecting each node.

Reliability and Fault Isolation

- **Star Topology:** Demonstrates superior fault isolation. An issue in a single cable or client device isolates that specific node without impacting the remainder of the network.
- **Bus Topology:** Possesses a single point of failure. Physical damage or improper signal termination along the main bus backbone brings down communication across all connected nodes.

Data Transmission Performance

- **Star Topology:** Offers consistent bandwidth and lower transmission collisions because data traffic routes through dedicated point-to-point connections managed by the central device.
- **Bus Topology:** Suffers severe throughput degradation as node count grows, resulting from increased signal collisions on the shared transmission medium.

5. Network Maintenance across Star, Mesh, Bus, and Hybrid Topologies

Maintenance, error isolation, and routine administration vary significantly depending on the underlying geometric arrangement of the network.

Star Topology Maintenance

- **Complexity Level:** Low
- **Diagnostics:** Centralized administration allows system engineers to immediately detect port errors and isolate broken client connections without affecting other users.

Mesh Topology Maintenance

- **Complexity Level:** High
- **Diagnostics:** Highly complex cable pathways and vast numbers of physical connections make manual hardware tracing and port auditing labor-intensive.

Bus Topology Maintenance

- **Complexity Level:** Very High
- **Diagnostics:** Finding a physical break or faulty terminator in a hidden cable run requires manual time-domain reflectometry or systematic line testing across the entire bus length.

Hybrid Topology Maintenance

- **Complexity Level:** Moderate to High
- **Diagnostics:** Faults are contained within individual sub-topologies, but ongoing management demands skilled administration familiar with multiple networking protocols and architectures.

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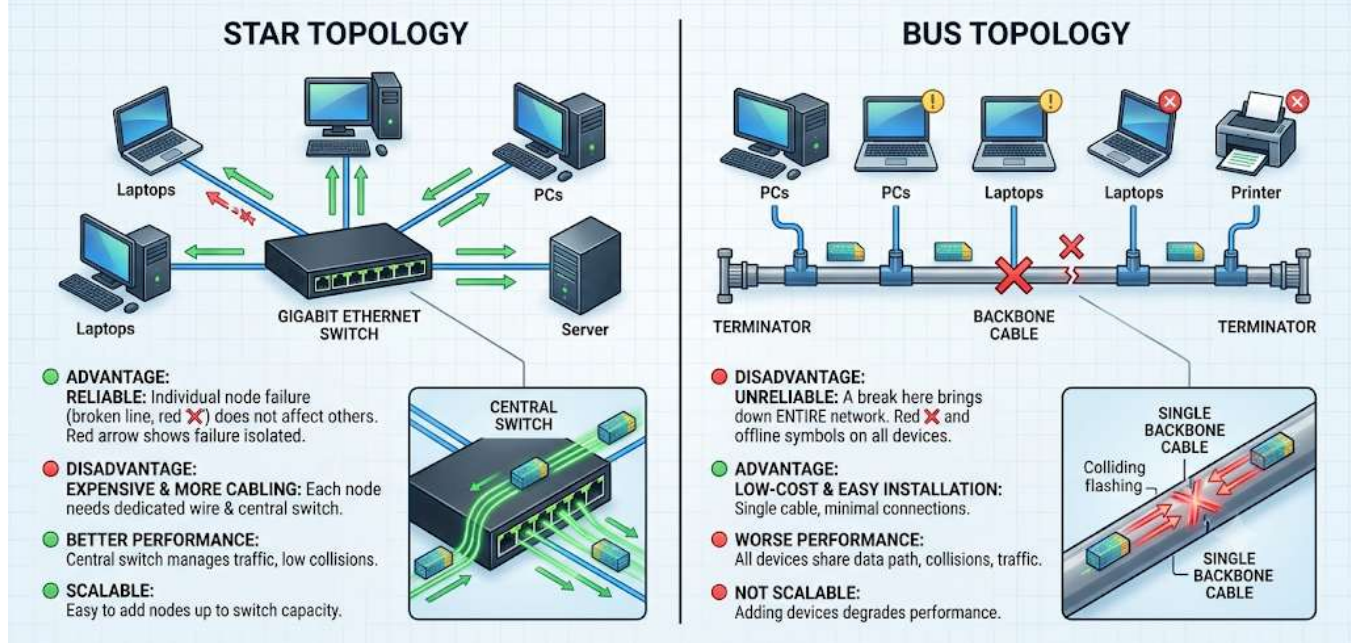
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NETWORK TOPOLOGY COMPARISON: STAR VS. BUS



2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

2. Mesh vs. Star Topology: Fault Tolerance and Redundancy

Network resiliency during hardware failures differentiates fully redundant mesh architectures from centralized star layouts.

Fault Tolerance Mechanics

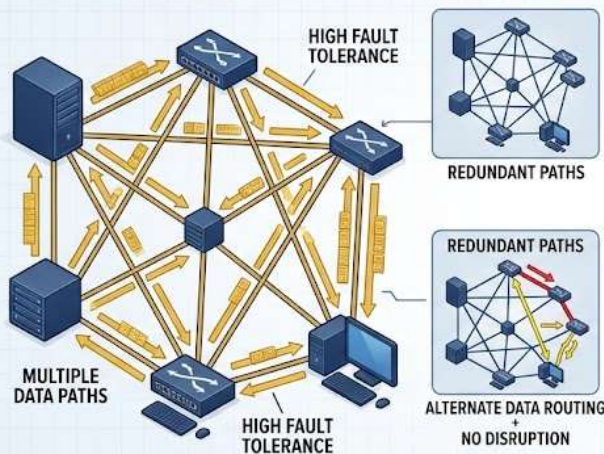
- **Mesh Topology:** Delivers maximum fault tolerance through redundant physical links. If any primary link breaks, routing protocols dynamically detour traffic through alternate paths without service disruption.
- **Star Topology:** Provides moderate fault tolerance. While end-node failures are contained safely, the central switch represents a critical single point of failure.

Application in High-Availability Environments

- **Mesh Topology:** Essential for mission-critical infrastructure (such as datacenter interconnections, defense networks, or industrial control systems) where downtime is unacceptable.
- **Star Topology:** Preferred for standard enterprise office networks where high availability is achieved by adding redundant switches rather than wiring every node to each other.

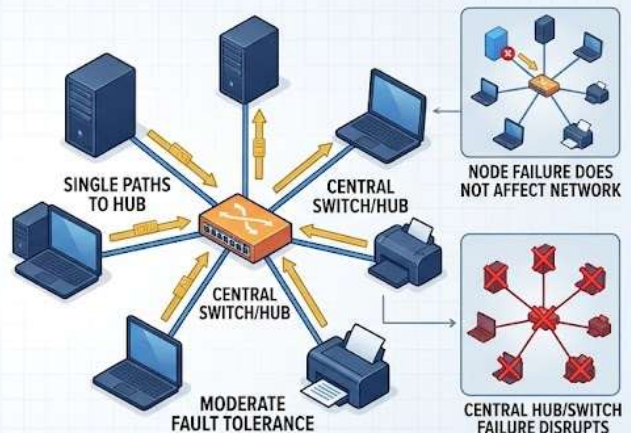
NETWORK TOPOLOGY COMPARISON: MESH VS. STAR

MESH TOPOLOGY (VERY HIGH FAULT TOLERANCE)



- EVERY NODE CONNECTED TO MULTIPLE NODES
- MANY PATHS FOR DATA TRANSMISSION
- ONE LINK FAILS, DATA STILL Routed (No disruption)
- HIGHLY ROBUST & RELIABLE (CRITICAL ENVIRONMENTS)

STAR TOPOLOGY (MODERATE FAULT TOLERANCE)



- ALL NODES CONNECTED TO CENTRAL SWITCH
- SINGLE PATHS VIA CENTRAL SWITCH
- FAILURE OF INDIVIDUAL NODES DOES NOT AFFECT NETWORK
- FAILURE OF CENTRAL SWITCH DISRUPTS ENTIRE NETWORK (Less robust)

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

3. Bus vs. Mesh Topology: Installation and Hardware Overhead

Physical installation complexity varies drastically between the minimalist structure of a bus network and the intricate wiring demands of a mesh network.

Deployment Complexity

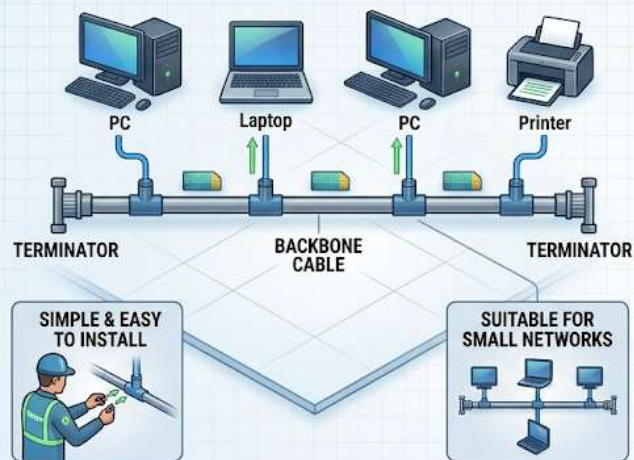
- **Bus Topology:** Simple and rapid to deploy in small, low-density settings using minimal cabling, taps, and basic connectors.
- **Mesh Topology:** Extremely complex to wire. A fully connected mesh network with N nodes requires $\frac{N(N-1)}{2}$ distinct physical links and dedicated network interface ports per machine.

Scaling Constraints

- **Bus Topology:** Straightforward initial connection, though adding too many endpoints creates signal attenuation and performance bottlenecks.
- **Mesh Topology:** Scaling becomes exponentially expensive and labor-intensive, requiring additional physical interface cards and massive cable routing for each added device.

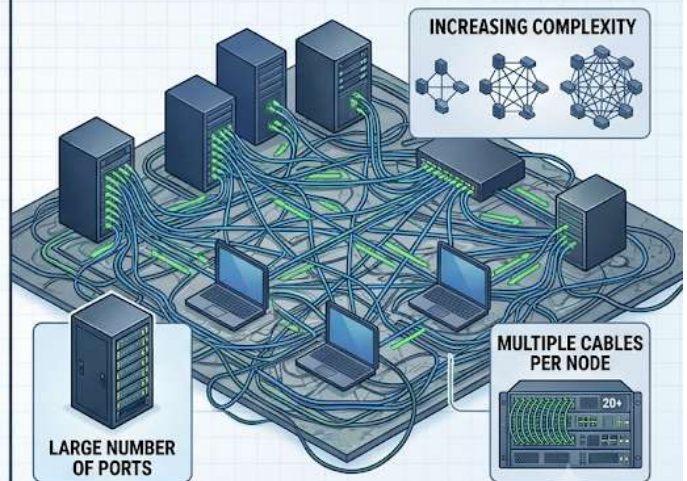
NETWORK TOPOLOGY COMPARISON: BUS VS. MESH

BUS TOPOLOGY (SIMPLE & LOW-COST INSTALLATION)



- Single backbone cable with minimal connections
- Straightforward to implement and configure
- Ideal for low-budget, small environments

MESH TOPOLOGY (HIGHLY COMPLEX & COSTLY INSTALLATION)



- Complex as each node connects to every other
- High hardware and maintenance overhead
- Time-consuming and costly implementation

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

4. Hybrid vs. Star Topology: Architectural Scalability

Scalability defines how efficiently a network framework accommodates long-term organizational expansion without demanding complete infrastructure overhauls.

Expansion Capabilities

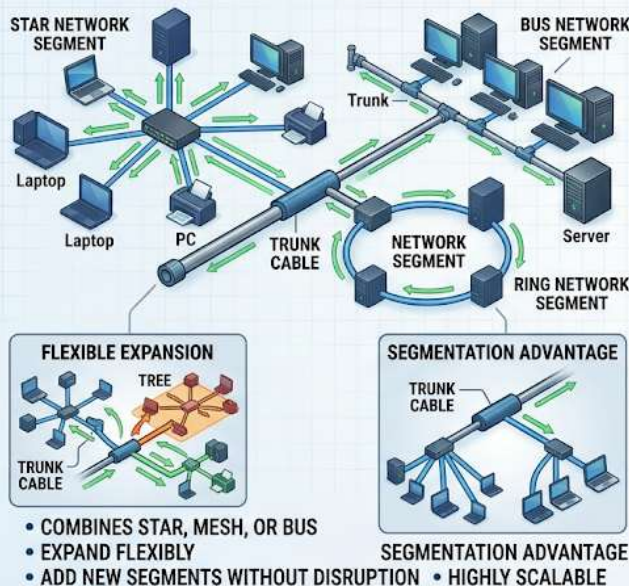
- **Hybrid Topology:** Combines two or more distinct topologies (e.g., Star-Ring or Star-Bus) into a single cohesive network. New operational units or entire branches can be added as standalone sub-networks without interrupting active segments.
- **Star Topology:** Scalability is constrained by the physical port density and processing capacity of the central switch or hub.

Structural Upgrades

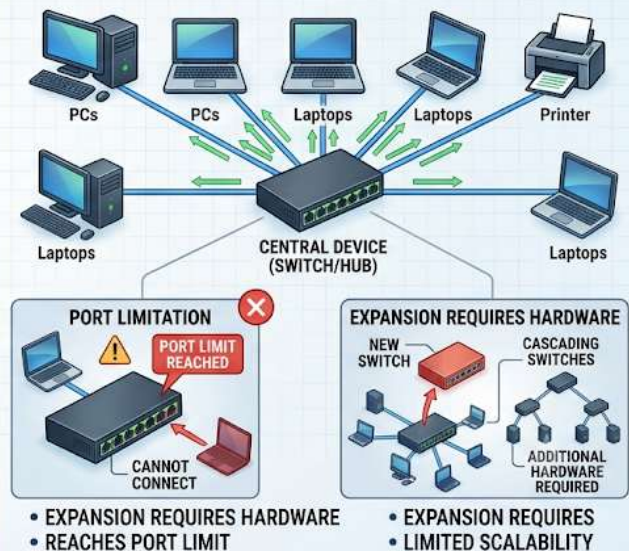
- **Hybrid Topology:** Allows tailored expansion using different topologies optimized for varying distances, physical layouts, or department requirements.
- **Star Topology:** Exceeding local port limits requires cascading additional switches or deploying high-density enterprise chassis switches.

NETWORK TOPOLOGY COMPARISON: HYBRID VS. STAR

HYBRID TOPOLOGY (HIGHLY SCALABLE)



STAR TOPOLOGY (LIMITED SCALABILITY)



- In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

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Bus Topology Maintenance

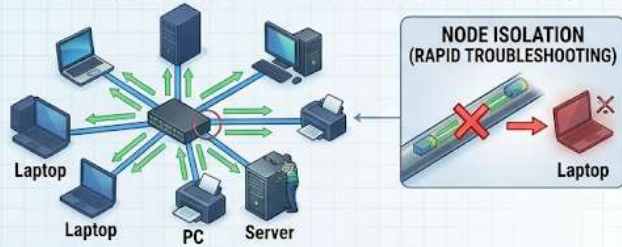
- **Complexity Level:** Very High
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Hybrid Topology Maintenance

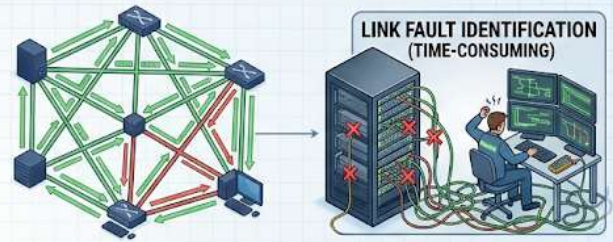
- **Complexity Level:** Moderate to High
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NETWORK TOPOLOGY COMPARISON: MAINTENANCE & TROUBLESHOOTING

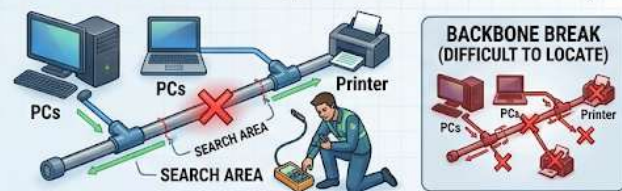
STAR TOPOLOGY (EASY TO MANAGE)



MESH TOPOLOGY (COMPLEX TO MAINTAIN)

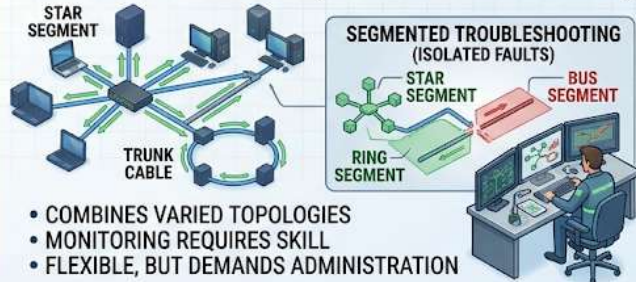


BUS TOPOLOGY (HARDER TO MAINTAIN)



- TECHNICIAN - FRUSTRATED DILEM
- FINDS TEST POINTS ALONG (E.G., TDR)
- MULTIPLE TEST POINTS DATA POINTS

HYBRID TOPOLOGY (MODERATE COMPLEXITY)



- COMBINES VARIED TOPOLOGIES
- MONITORING REQUIRES SKILL
- FLEXIBLE, BUT DEMANDS ADMINISTRATION

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion